Q1 - Stereo hearing. Stereo systems often use a single subwoofer that plays low-frequency sounds. The reasoning is that we don't hear the direction of bass sounds, so there's no point using two subwoofers. Below what frequency will we struggle to tell the direction of a sound? Assume the human head is 25cm across and our brain can detect when two signals are a radian out of phase.

The maximum difference in path length between our ears is the width of our head, or 25cm. If we define that to be 1 radian, then the wavelength must be $25 \times 2\pi = 157$ cm, or 1.57m. Sound with a wavelength of 1.57 metres has a frequency of $340/1.57 = 216$ Hz. We would expect to struggle to tell direction once we're well below about 200 Hz. Indeed, wikipedia says this is about the top frequency of consumer-grade subwoofers. Professional-grade systems usually have a lower subwoofer frequency, presumably because the main speakers to a better job with bass, and your sense of directionality rolls off gradually.

Q2 - More localization. We've talked about time delays for localizing sounds, but our brain uses several other clues, including loudness. If a sound is on one side of our head then our head blocks the sound from the ear on the far side from the sound. You might think this would let you localize bass, but in fact you still can't. Please explain why, and estimate the frequency below which this won't work. You may assume your head has not changed size since the previous problem.

If our head is much smaller than a wavelength, then the sound will just go around our heads without getting blocked. If we want to be quantitative, we can take the wavelength at which the bending angle is 90 degrees, since that means a wave will wrap directly around our heads. If we use $\sin(\theta) = \lambda/d$, then for a 90 degree angle, $sin(\theta)$=1. We know that d is 25 cm, so we have

$$1 = \frac{\lambda}{d} \tag{1}$$

or $\lambda = d$. That gives a wavelength of 25 cm, and a frequency of 1360 Hz. Reasonable people could disagree on factors of 2, so anything in this range (as long as it's justified!) would be acceptable.

Q3 - Two cardinals are sitting in a tree 20m away. The cardinals are 0.5m apart. What is the angle separating the cardinals, assuming the branch they're sitting in is perpendicular to your line of sight?
One of the cardinals starts singing. Assume their song has lots of notes that go up to 3.4 kHz. Without looking, could you tell which one is singing?

If we point our faces towards one of the cardinals, the path length difference to our ears is zero. The path length difference to the other cardinal is then $d\sin(\theta)$ where d is the size of our head, 25 cm. We know that $\tan(\theta) = 0.5/20 = 0.025$. That qualifies as a small angle, so $\tan(\theta) = \sin(\theta)/\cos(\theta) \simeq \theta/1 = \theta$. For small angles, $\tan(\theta)$ is also approximately equal to θ, so we take $\theta = 0.025$. The path length difference is $0.025 \times 25cm = 0.625cm = 6.2e-3m$. If we say our ears can hear a difference of a radian, then the wavelength would be 2π times the path length difference, or about 4 cm/0.04m. To convert to frequency, we take $v/\lambda = 340/0.04 = 8500$ Hz or 8.5 kHz. I said the cardinal's top frequency was 3.4 kHz, so we would struggle to tell which one was singing. We aren't far away, though, so with a bit more high frequency singing we could probably

tell which bird is singing.

Q4 - Modes of a room. You find yourself in a square room that is 6.8m on a side and 3.4m tall.
A) What are the fundamental frequencies of the room along the principle directions? (i.e. length/width (which will be the same) and height)

If we have a closed room with side 6.8m, then the wavelength of the fundamental is twice that, or 13.6m. 340/13.6=25 Hz, so the fundamental in the horizontal directions is 25 Hz. The room height is 3.4m, double that to get a wavelength of 6.8m, which corresponds to 50 Hz.

B) Remember that the frequencies of a rectangular room are $\sqrt{m_x^2\nu_x^2 + m_y^2\nu_y^2 + m_z^2\nu_z^2}$ where ν_x, ν_y, ν_z are the fundamental frequencies in each direction and m_x, m_y, m_z are arbitrary non-negative integers. If we want to find all resonant frequencies of the room *below* 100 Hz (so not including 100 Hz), what values can each of m_x, m_y, and m_z have?

If we're cutting out 100 Hz, then the horizontal directions can have frequencies of 0, 25, 50, or 75 Hz or m_x and m_y can be $0, 1, 2, 3$. The vertical direction can have frequencies of 0 or 50 Hz, so m_z can be either 0 or 1.

C) For those who feel like a challenge, how many *distinct* resonant frequencies will our room have below 100 Hz, and what are the corresponding values of m_x, m_y, and m_z?

If we swap m_x and m_y the frequency doesn't change, so we can declare that m_y is always at least as big as m_x (if m_y were smaller, we can just swap m_x and m_y). So we want all ordered pairs (m_x, m_y, m_z) where m_x is between 0 and 3, m_y is at least as big as m_x, and m_z is either 0 or 1.

We'll start with $m_z = 1$, so the ordered pairs where this works are: (0,0,1), (0,1,1), (0,2,1), (0,3,1), (1,1,1), (1,2,1),(1,3,1),(2,2,1), (2,3,1),(3,3,1). We can repeat with $m_z = 0$, except we can't have $m_x = m_y = m_z = 0$ since that would be a zero frequency mode. We just take the same set of numbers we had before, except we flip m_z from 1 to 0, and skip the first entry since that would now be all zeros. We then get (0,1,0),(0,2,0),(0,3,0),(1,1,0),(1,2,0),(1,3,0),(2,2,0),(2,3,0),(3,3,0). That's 19 pairs for (m_x, m_y, m_z). Some of these give the same frequency, though. We want to find

$$\sqrt{25^2 m_x^2 + 25^2 m_y^2 + 50^2 m_z^2} \tag{2}$$

$$= \sqrt{25^2 m_x^2 + 25^2 m_y^2 + (2 * 25)^2 m_z^2} \tag{3}$$

$$= 25\sqrt{m_x^2 + m_y^2 + 4m_z^2} \tag{4}$$

If two modes have the same value for $m_x^2 + m_y^2 + 4m_z^2$ then they'll have the same resonant frequency.

I'll list all the modes and their corresponding values of $m_x^2 + m_y^2 + 4m_z^2$:

$$(0,1,0) \rightarrow 1 \tag{5}$$
$$(0,2,0) \rightarrow 4 \tag{6}$$
$$(0,3,0) \rightarrow 9 \tag{7}$$
$$(1,1,0) \rightarrow 2 \tag{8}$$
$$(1,2,0) \rightarrow 5 \tag{9}$$
$$(1,3,0) \rightarrow 10 \tag{10}$$
$$(2,2,0) \rightarrow 8 \tag{11}$$
$$(2,3,0) \rightarrow 13 \tag{12}$$
$$(3,3,0) \rightarrow 18 \tag{13}$$
$$(0,0,1) \rightarrow 4 \tag{14}$$
$$(0,1,1) \rightarrow 5 \tag{15}$$
$$(0,2,1) \rightarrow 8 \tag{16}$$
$$(0,3,1) \rightarrow 13 \tag{17}$$
$$(1,1,1) \rightarrow 6 \tag{18}$$
$$(1,2,1) \rightarrow 9 \tag{19}$$
$$(1,3,1) \rightarrow 14 \tag{20}$$
$$(2,2,1) \rightarrow 12 \tag{21}$$
$$(2,3,1) \rightarrow 17 \tag{22}$$
$$(3,3,1) \rightarrow 22 \tag{23}$$
$$\tag{24}$$

If we go through the values in Equation **??**, we get unique values of (1,2,4,5,6,8,9,10,12,13,14,17,18,22) or 14 distinct frequencies. Once you take the square root of those values, that means you get 13 overtones between the fundamental and 5 times the fundamental. Most of them aren't integer ratios (you get an octave and a twelfth, but no others), so if this room vibrates it won't sound like a clean pitch.

Q5 - Mode spacing on a string. Assume you have a string clamped at both ends that has n harmonics below some cutoff frequency ν_{cut}. How many harmonics will there be below $2\nu_{cut}$?

We have harmoncs spaced by the fundamental ν_f, so as long as ν_{cut} is above ν_f, then we'll get ν_{cut}/ν_f harmonics. If we double the cutoff frequency, we get $2\nu_{cut}/\nu_f$. Divide those two to get the ratio and we get:

$$\frac{2\nu_{cut}/\nu_f}{\nu_{cut}/\nu_f} = 2 \tag{25}$$

So, ignoring rounding effects, we will get twice as many harmonics by going up a factor of 2 in frequency.

Q6 - Mode spaing in a room. Redo problem 5, but for a roughly (but not exactly) cubical

room. Hint - if m_x, m_y, and m_z are less than some value m_{cut}, then the total number of modes is roughly m_{cut}^3 because there are m_{cut} possibilities for each of m_x, m_y and m_z.

The max value of m_x if the max value where $m_x\nu_x < \nu_{cut}$ or

$$m_x < \frac{\nu_{cut}}{\nu_x} \tag{26}$$

and similarly for m_y and m_z. The max value for m_x is then $m_{x,max} = \frac{\nu_{cut}}{\nu_x}$. If the room dimensions aren't integer periods of each other, we don't have to worry about repeated frequencies, and so the number of modes we have is a correction factor f times the number of values for $m_{x,y,z,max}$. Since we can pick each of those values independently, the total number of ways we have to pick is $fm_{x,max}m_{y,max}m_{z,max}$. If we double the cutoff frequency, we double the max m values, but the correction factor should stay roughly the same. That means the new number of modes is

$$f2m_{x,max}2m_{z,max}2m_{z,max} = 8fm_{x,max}m_{y,max}m_{z,max} \tag{27}$$

That is exactly the number of modes we had before, except for that factor of 2^3 or 8. More generally, if we have an object vibrating in n dimensions, the number of modes below a cutoff frequency scales like ν_{cut}^n. If we want a pitch, we want the frequencies to line up, so we want a one-dimensional object vibrating. Pitched instruments are long and skinny (what about string instruments? Well, the part that's actually vibrating is the string, which is indeed very long and skinny). Percussion instruments where we want lots of modes are going to be more round.

Discussion of Q5 and Q6: You should find that if we double the cutoff frequency on a string you get twice as many modes. That means the number of modes between 0 and ν_{cut} is the same as the number of modes between ν_{cut} and $2\nu_{cut}$. The average *separation* between the frequencies of the harmonics is the same. For the room, you should find there are eight times as many modes going to $2\nu_{cut}$, so the number of frequencies between ν_{cut} and $2\nu_{cut}$ is seven times as many as between 0 and ν_{cut}. The separation between mode frequencies gets smaller as we go to higher frequency, and eventually they will get so closely spaced that the room no longer acts like a resonator with distinct frequencies.